

Actual Euclidean certificates in a patterned AQ subfamily

Hruday N M (BUNZEEY)

22 September 2026

Round31 scientific addendum: AT4–AT6

Abstract

We give analytic enclosures for the actual centered Wilson correlation in the zero-selected, nonuniform AQ lattice family, retaining its chosen infinite-volume subsequential state and physical energy generator. A local reset estimate, seven-star bounded-perturbation comparison and positive Poisson integration yield a complete error budget. At the initially fixed coupling $\tau = 10^{-8}$, that budget misses the 10^{-6} point-error target. After this reviewed failure, a prospectively declared smaller nonzero coupling $\tau = 10^{-14}$ yields a certified datum at $s = 1$. A third investigation keeps that coupling and state fixed, proves a continuous bound on $0 \leq s \leq 128$, and exports all 4097 rational proxies with actual-correlation error at most 10^{-6} . The resulting reduced inverse-energy form $I = \alpha\mathcal{R}$ has a certified interval of width below $1/500$, improved below $1/2500$ by paying the actual endpoint tail. The stronger displayed enclosure is $I \in [0.083174562675757, 0.083535946641292]$. It contains the free value $1/12$: no interaction-induced shift is resolved. These are analytic actual-model bounds, not measured or simulated interacting trajectories. They establish neither uniform Wilson theory nor a continuum Yang–Mills construction.

Status and reading guide

This is a separate addendum to the unchanged Draft03 checkpoint. Exactly three prospectively contracted investigations are complete. AT4 is limited at its frozen precision target; AT5 and AT6 are accepted within their stated model scope. Sections 2–4 derive the certificates and explain the adaptive change between investigations. The final sections state source attribution, remaining goals, and the reproducibility boundary.

The human project author is Hruday N M (BUNZEEY). Forward, reverse and skeptical work used separate model-agent roles with shared mathematical ancestry. This is neither external human peer review nor independent physical observation. Historical figures supply documented methodological lenses; none participated in or endorsed this work. HNM contribution aliases identify project records; scientific priority is unverified.

1 Model, inherited premises and fixed physical clock

This addendum starts from the reviewed AQ construction of Round29 and the same-state Wilson estimates of Round30. Its compactness/GNS construction retains the prior-strategy attribution to Gauvin’s Supplement A.10, with the actual AQ operator and dynamics dictionary checked separately [2, 4, 1]. This imports neither that source’s different-model constants nor its larger physical claims. The theory here is at fixed lattice spacing. A coarse site $b \in \mathbb{Z}^3$ owns all three positive links from each fine tail $(4b_x + r, 2b_y + q, b_z)$, $0 \leq r < 4$, $0 \leq q < 2$: exactly 24 untruncated $L^2(\text{SU}(2))$ link

factors. Both original endpoint gauge actions are retained. Write $C_e = -\sum_a X_{ea}^2$, with generators $i\sigma_a/2$, and $W_f = \frac{1}{2} \text{Tr } U_f$ for the real fundamental Wilson plaquette.

The selected strip coefficients obey

$$|\lambda_L|, |\lambda_R| \leq \alpha/2, \quad |\mu_M| \leq \alpha/8. \quad (1)$$

Round31 selects $(\lambda_L, \mu_M, \lambda_R) = (0, 0, 0)$. This is an admissible closed-box parameter choice, not a replacement of a nonzero selected vacuum by Haar. The 21 remaining anchored face classes have the nonzero coefficient $\nu = \alpha\tau/24$. Whole stars $S = \{0, e_x, e_y, e_z\}$ are retained only when $b + S \subset \Lambda_N$, where $\Lambda_N = [-N, N]^3$. The raw finite Hamiltonian is

$$H_{N,\tau}^{\text{raw}} = \alpha \sum_{e \text{ owned by } \Lambda_N} C_e - \frac{\alpha\tau}{24} \sum_{b+S \subset \Lambda_N} \sum_{f \in O_b} W_f, \quad |O_b| = 21. \quad (2)$$

Its actual ground energy is subtracted before spectral statements. A magnetic additive scalar, an on-site reference ground shift and the actual interacting ground energy are different objects; scalar cancellation in a conjugation does not license omitting ground centering in a correlation.

This is a *patterned, nonuniform magnetic model*. Three selected xy classes have zero coefficient. Equality with the uniform Kogut–Susskind model would require $\lambda_L = \mu_M = \lambda_R = \nu$, which fails for $\tau \neq 0$ here. The original AL dictionary $\tau = 96/g^4$ for equal coefficients is not assigned to this subfamily. Small τ does not establish a weak-bare-coupling or continuum trajectory.

The positive energy α , action $\hbar$, lattice spacing a and comparison energy $E_\star$ remain fixed. AQ uses $\delta = \alpha/8$. Let ω_τ be any of the centered-box subsequential states supplied by the AQ construction for the declared coefficients, and let $H_\tau \geq 0$ be its actual physical GNS energy generator, with vacuum Ω_τ . The inherited gap in that representation is at least $\alpha/16$. No uniqueness of all thermodynamic states is assumed. Both the reference and interacting clocks use

$$K_\tau = H_\tau/\alpha, \quad s = \frac{\alpha t_E}{\hbar}, \quad t = \frac{\alpha t_{\text{real}}}{\hbar}. \quad (3)$$

Thus s, t, τ are dimensionless, while a reduced inverse-energy form has units $1/\alpha$. A real-time cutoff L is a proof resource, not a physical deformation. Changing τ between prospectively frozen investigations is an explicit interaction change, not a redefinition of units.

For the original xz Wilson plaquette at the origin set

$$m_\tau = \omega_\tau(W), \quad \chi_\tau = (\pi_\tau(W) - m_\tau)\Omega_\tau, \quad C_\tau(s) = \langle \chi_\tau, e^{-sK_\tau} \chi_\tau \rangle. \quad (4)$$

It is dimensionless and centered in the actual state. Its complete-factor cover is $R = \{0, e_z\}$: 48 stored links and 36 original endpoints, with seven incident whole-star anchors

$$R - S = \{0, -e_x, -e_y, -e_z, e_z, e_z - e_x, e_z - e_y\}. \quad (5)$$

This seven-star bulk region must not be replaced by the two-star orthant configuration. The on-site product reference at the chosen zero selected triple is Haar; its single-plaquette vector has variance $1/4$ and energy 3α . Those facts describe the reference, not an exact interacting spectrum.

2 AT4: a direct actual-state comparison and its failed precision target

Contribution HNM-C-AT4. The first contract fixed $\tau = 10^{-8}$, $s = 1$, $L = 10^4$ and absolute point error 10^{-6} . Its reviewed verdict is *limited*: an actual-state interval is obtained, but it fails the specified precision. The following derivation records the sharper forward certificate; the reverse proof independently supplied a valid broader tail bound. The effect-bound refinement was separately checked after both reports froze and is not claimed as a duplicated independent discovery.

2.1 Local state error

At the zero selected triple the dimensionless on-site reference is $h_b = 8 \sum_{e \text{ owned by } b} C_e$. Its first positive full-space energy is $8(3/4) = 6$. Therefore, with the Haar product projection P_R ,

$$h_R \geq 6(I - P_R). \quad (6)$$

Reset the finite interacting ground density on R to P_R , preserving its exterior marginal. Each of the seven incident whole stars changes its expectation by at most $2(7|\tau|)$ in AQ normalized units. The variational principle and cancellation of unchanged exterior energy give

$$\text{Tr}(\rho_{N,R} h_R) \leq 98|\tau|. \quad (7)$$

Positive spectral truncations justify the cancellation for unbounded reference operators. The reset trial and finite ground have finite form energy because the finite interaction is bounded. Pass bounded truncations through the actual AQ local trace-norm limit, then take their monotone supremum. Thus

$$1 - \text{Tr}(\rho_{\tau,R} P_R) \leq \frac{49|\tau|}{3}, \quad \|\rho_{\tau,R} - P_R\|_1 \leq D := 2\sqrt{\frac{49|\tau|}{3}}. \quad (8)$$

The second inequality follows by pure-state trace distance, convexity and Cauchy–Schwarz. It is uniform for the AQ subsequential states under this construction; it does not identify them. No convergence-rate assertion is used.

Trace duality gives $|m_\tau| \leq D$. A trace-zero density difference of norm at most D has expectation at most $D/2$ against an effect. Applying this to $0 \leq W^2 \leq I$ gives

$$\left| \omega_\tau(W^2) - \frac{1}{4} \right| \leq D/2, \quad v_\tau := \|\chi_\tau\|^2 \leq \frac{1}{4} + D/2. \quad (9)$$

The complex operator $W\alpha_t^0(W)$ below is not an effect; its error remains D , not $D/2$.

2.2 Dynamics and centering

In each finite box split $H_{N,\tau}^{\text{raw}}/\alpha = A_N + B_N$, where A_N contains free evolution on R , all outside on-site terms and every outside interaction disjoint from R . Put all seven incident star groups into B_N . Each such star has norm at most $7|\tau|/8$ in the physical/alpha convention, so

$$\|B_N\| \leq 49|\tau|/8, \quad \left\| \alpha_t^\tau(W) - \alpha_t^0(W) \right\| \leq k|t|, \quad k := 49|\tau|/4. \quad (10)$$

To justify the estimate, differentiate the relative unitary $e^{it(A_N+B_N)}e^{-itA_N}$ on the common operator domain. The derivative contains only the bounded perturbation B_N ; strong integration extends the identity to every vector and bounds the relative unitary's distance from I by $|t| \|B_N\|$. Insert bounded W only afterwards to obtain the factor two in the conjugation bound. This requires no norm derivative of arbitrary local bounded operators. AQ's already matched unbounded-on-site dynamics theorem [4, 1] passes the uniform finite-volume bound to the actual limiting dynamics. Replacing seven stars by the extensive magnetic norm would lose this volume uniformity.

The free evolution preserves R and $c_0(t) = \frac{1}{4}e^{3it}$. Add and subtract the free evolution in the actual state, then its Haar expectation, keeping the actual mean subtraction:

$$c_\tau(t) = \omega_\tau(W\alpha_t^\tau(W)) - m_\tau^2, \quad (11)$$

$$|c_\tau(t) - c_0(t)| \leq k|t| + D + D^2, \quad (12)$$

$$|c_\tau(t) - c_0(t)| \leq v_\tau + \frac{1}{4} \leq B := \frac{1}{2} + D/2. \quad (13)$$

The first bound charges dynamics, local state and centering separately; the second provides a bounded whole-tail allowance.

2.3 Poisson transfer and evaluated certificate

The established Fourier identity [3]

$$\int_{\mathbb{R}} \frac{s e^{itx}}{\pi(s^2 + t^2)} dt = e^{-s|x|} \quad (s > 0) \quad (14)$$

becomes a heat identity only because the actual ground-centered GNS spectrum is nonnegative. The positive kernel has mass one and the spectral measure has finite mass, so spectral Fubini is legitimate. Split the integral at $|t| = L$. Use (12) inside and (13) on the entire complement. The two elementary integrals are

$$\int_{-L}^L \frac{s|t|}{\pi(s^2 + t^2)} dt = \frac{s}{\pi} \log(1 + L^2/s^2), \quad (15)$$

$$\int_{|t|>L} \frac{s}{\pi(s^2 + t^2)} dt = \frac{2}{\pi} \arctan(s/L) \leq \frac{2s}{\pi L}. \quad (16)$$

Consequently the actual AQ correlation obeys

$$\left| C_{\tau}(s) - \frac{1}{4}e^{-3s} \right| \leq \mathcal{E}_{\tau}(s, L) := D + D^2 + \frac{ks}{\pi} \log(1 + L^2/s^2) + B \frac{2s}{\pi L}. \quad (17)$$

Increasing L reduces the tail but increases the logarithmic interior term; neither may be silently discarded. The formula is a model-specific sufficient error bound, not a new generic Poisson or Duhamel theorem.

At the frozen AT4 parameters the outward-certified forward error allowance is approximately 0.000841518704386267, with the components in Table 1. The actual interval is approximately [0.0116052483875797, 0.0132882857963523]. The exact rational endpoints and arithmetic proof are in the [frozen AT4 output](#). Its midpoint radius exceeds 10^{-6} , so it is not a node satisfying AT3’s input contract. This is failure of this upper certificate; it is not a lower bound on the true interacting-minus-reference error.

Table 1: AT4 forward budget; decimals are readable summaries of outward rational bounds.

Cost	Certified allowance, approximately
Local state D	$8.08290376865476 \times 10^{-4}$
Mean-square centering D^2	$6.53333333333333 \times 10^{-7}$
Interior dynamics	$7.18276887291769 \times 10^{-7}$
Whole Poisson tail	$3.18567173001654 \times 10^{-5}$
Total	$8.41518704386267 \times 10^{-4}$

2.4 Information and damaging controls

The inherited abstract measures $\eta_A = (\delta_2 + \delta_4)/8$ and $\eta_B = \delta_1/32 + 3\delta_3/16 + \delta_5/32$ have common mass, first moment and second moment $(1/4, 3/4, 5/2)$. Writing $r = e^{-1}$, their nonzero-time difference is exactly

$$C_B(1) - C_A(1) = \frac{r(1-r)^4}{32} > 0. \quad (18)$$

Every deterministic estimator using only those common moments returns the same number on both. The triangle inequality therefore gives a worst-case error of at least half their separation, strictly greater than 9/10000 by directed exponential arithmetic. This information limit concerns an abstract admissible moment class, not multiple AQ realizations or nonuniqueness of the actual state.

The calculators accept exact rational inputs and enforce the zero selected triple, $|\tau| \leq 10^{-8}$, positive time/cutoff/scales and their frozen-design constraints. Binary floats, booleans, malformed/nonfinite values and model deformations are rejected. Admissible inputs with a broad radius return an explicit insufficient result. Both coupling signs receive the same absolute certificate; the exact $\tau = 0$ branch uses its separate free identity. Keeping the old cutoff at the new coupling fails because of its tail; keeping the old coupling at the new cutoff fails because of its state term. Neither failure is concealed.

Directed arithmetic uses integer square-root brackets, alternating Taylor enclosures with argument reduction and positive squaring, positive logarithm series with an explicit remainder, and Machin/arctangent bounds for π . Every admission decision is rational arithmetic, and the exponential midpoint’s own error is included in (19). Separate normal and optimized producer runs reproduce their frozen outputs. The reviewer audited the APIs, domain mutations, radius sums and preservation of the free reference.

Full derivations and reviewed implementation: [AT5 forward](#), [AT5 reverse](#), and [independent model-agent review](#). AT5 by itself certifies one node; it does not evaluate a time grid or an inverse response. The next section records the separately selected third investigation.

4 AT6: a complete actual-state grid and inverse-energy enclosure

Contribution HNM-C-AT6. The third contract retains AT5’s same chosen actual state, selected triple $(0, 0, 0)$ and $\tau = 10^{-14}$. It fixes

$$T = 128, \quad h = 1/32, \quad N = 4096, \quad s_j = jh, \quad L = 10^9, \quad \epsilon = 10^{-6}. \quad (23)$$

Thus the physical times are $t_{E,j} = j\hbar/(32\alpha)$ and the final time is $128\hbar/\alpha$. Neither the state, coupling nor clock changes across the nodes. The accepted result supplies the previously missing actual-model input premise only within this specified small-coupling family.

4.1 The entire window, including zero

For $s > 0$, use the complete error $\mathcal{E}_\tau(s, L)$ in (17). Put $f(s) = s \log(1 + (L/s)^2)$ and $y = L/s$. For every real $0 < s \leq T$, $y \geq L/T > 3$, and

$$f'(s) = \log(1 + y^2) - \frac{2y^2}{1 + y^2} > \log 10 - 2 > 0. \quad (24)$$

The final inequality is checked by an exact lower enclosure $e^{-2} > 1/10$. The other time-dependent cost is positive and linear, so the bound increases throughout the positive window. This is a continuous proof, not a conclusion from the finite grid.

At zero, use the actual variance separately. Equation (9) and $m_\tau^2 \leq D^2$ give

$$|C_\tau(0) - 1/4| \leq D/2 + D^2. \quad (25)$$

The singular Poisson kernel is not evaluated at zero. Outward rational evaluation at T gives a positive-time allowance E_{window}^+ satisfying

$$D^+/2 + (D^+)^2 < E_{\text{window}}^+ < 8.491932 \times 10^{-7} < \epsilon. \quad (26)$$

The zero-time allowance is approximately $4.041458418 \times 10^{-7}$. State replacement, mean-square centering, dynamics and the complete Poisson tail are all retained. The reverse implementation keeps its conservative tail and proves the separate uniform bound below 9.714242×10^{-7} , also below ϵ .

4.2 Every exported node has a paid actual error

The forward arithmetic uses $G = 10^{30}$ and an outward rational enclosure $q_- \leq e^{-3/32} \leq q_+$. Starting from $p_0^- = p_0^+ = 1$, it computes

$$p_{j+1}^- = \frac{\lfloor G p_j^- q_- \rfloor}{G}, \quad p_{j+1}^+ = \frac{\lceil G p_j^+ q_+ \rceil}{G}. \quad (27)$$

Positive multiplication and induction enclose $e^{-3j/32}$ at every node. With $f_j^\pm = p_j^\pm/4$, define

$$d_j = (f_j^- + f_j^+)/2, \quad a_j = (f_j^+ - f_j^-)/2, \quad (28)$$

$$b_0 = D^+/2 + (D^+)^2, \quad b_j = E_{\text{window}}^+ \quad (j > 0), \quad (29)$$

$$|C_\tau(s_j) - d_j| \leq e_j := b_j + a_j \leq \epsilon. \quad (30)$$

The largest arithmetic half-width is exactly $3/(2 \cdot 10^{30})$; the complete maximum node error remains below 8.491932×10^{-7} . The first reference bin is exactly $[1/4, 1/4]$, but its actual-state error is positive. The final reference bin is $[0, 11/(4 \cdot 10^{30})]$, with exact datum $11/(8 \cdot 10^{30})$. Its zero lower endpoint does not declare the exponential to be zero. The reverse recurrence uses $G = 10^{36}$ and pays its own complete arithmetic and analytic errors.

All 4097 rows are exported with exact indices, times, reference bins, data and complete actual-error allowances. They are rational reference proxies whose distance from the actual interacting correlation has been proved. They are neither observations nor simulated interacting trajectories. The theorem and frozen source packet establish provenance; a metadata label on arbitrary data does not do so.

4.3 The same centered spectral measure and its two tails

Let η be the spectral measure of χ_τ in $x = E/\alpha$. The inherited same-state AQ/AT results give

$$\text{supp } \eta \subset [g, \infty), \quad g = 1/16, \quad \eta(\mathbb{R}) \leq M = 63/250, \quad \int x d\eta \leq U = 94/125, \quad (31)$$

and $\int x^2 d\eta \leq 36 + 98|\tau|$. The gap and positive measure imply by Tonelli

$$I := \int_0^\infty C_\tau(s) ds = \int x^{-1} d\eta(x) = \alpha \mathcal{R}, \quad \mathcal{R} := \langle \chi_\tau, (H_\tau|_{\Omega_\tau^\pm})^{-1} \chi_\tau \rangle. \quad (32)$$

The notation $\mathcal{R}$ distinguishes this inverse-energy form from the local support R . The spectral measure and centering remain the same in every step.

The historical mass bound pays

$$0 \leq \int_T^\infty C_\tau(s) ds \leq D_{\text{mass}} := (M/g)e^{-gT} = \frac{504}{125}e^{-8}. \quad (33)$$

The prospectively shared endpoint candidate is also a standard spectral consequence: $x \geq g$ implies $e^{-Tx}/x \leq e^{-Tx}/g$. Integrate this pointwise inequality against the same positive η to obtain

$$\int_T^\infty C_\tau(s) ds \leq C_\tau(T)/g = 16C_\tau(T) \leq 16u_T, \quad u_T := \max(0, d_N + e_N). \quad (34)$$

The forward value $16u_T$ is approximately $1.358709054 \times 10^{-5}$, versus mass-tail allowance approximately 0.001352585316 . Use $D_{\text{new}} = \min(D_{\text{mass}}^+, 16u_T)$, where the superscript denotes the certified rational upper evaluation. The tiny free endpoint alone is not an actual tail bound. A constant finite-window comparison error is never integrated to infinite time.

4.4 Quadrature sign, deterministic errors and evaluated intervals

The finite second moment justifies $C''_\tau(s) = \int x^2 e^{-sx} d\eta(x)$, including the right derivative at zero. Positivity and Tonelli yield $\int_0^T C''_\tau(s) ds \leq U$. On cell $[jh, (j+1)h]$, twice integrating by parts gives trapezoid minus integral as the integral of C''_τ against the nonnegative Peano kernel $(s-jh)((j+1)h-s)/2 \leq h^2/8$. Consequently

$$0 \leq \text{Trap}(C_\tau) - \int_0^T C_\tau(s) ds \leq Q := h^2 U / 8 = 47/512000. \quad (35)$$

Every exported datum is used with the actual trapezoid weights:

$$P = \frac{1}{32} \left(\frac{d_0}{2} + \sum_{j=1}^{4095} d_j + \frac{d_{4096}}{2} \right), \quad |P - \text{Trap}(C_\tau)| \leq J := T\epsilon = 2/15625. \quad (36)$$

The weights sum to T . Same-sign errors attain $T\epsilon$, so a statistical square-root reduction is invalid. No geometric closed sum replaces the exported-node readout.

The unchanged historical AT3 evaluator receives every exact datum as the bin $[d_j, d_j]$. Its extra arithmetic-bin width is zero because a_j is already charged in e_j . Its conditional and uncomputed-sample flags remain unchanged; the new wrapper discharges the error premise for this source-bound construction. It does not rewrite the historical result or turn the proxies into measured data. The two admitted inverse intervals are

$$I \in [\max(0, P - Q - J), P + D_{\text{mass}}^+ + J], \quad (37)$$

$$I \in [\max(0, P - Q - J), P + D_{\text{new}} + J]. \quad (38)$$

The quadrature allowance lowers the lower endpoint. Nonnegative clipping is justified by $I \geq 0$ and is inactive here. Both deterministic costs and the actual endpoint error are paid even when correlated.

Table 3: AT6: short outward rational enclosures for $I = \alpha\mathcal{R}$. The displayed terminating decimals are exact widened endpoints.

Readout	Enclosing interval	Target met
Forward, mass tail	[0.083174562675757, 0.084874944866461]	$< 1/500$
Forward, endpoint	[0.083174562675757, 0.083535946641292]	$< 1/2500$
Reverse, endpoint	[0.083174562675757, 0.083537902336706]	$< 1/2500$

For the forward output, the exact lower endpoint and trapezoid are

$$L_I = \frac{42585376089987669271345664867041}{512000000000000000000000000000000}, \quad (39)$$

$$P = \frac{42697912089987669271345664867041}{5120000000000000000000000000000}. \quad (40)$$

The full rational upper endpoints and tail costs are exported in the [forward results](#) and [reverse inverse ledger](#). Before the short decimal widening in Table 3, the exact forward widths are $Q + D_{\text{mass}}^+ + 2J < 0.001700382190703$ and $Q + D_{\text{new}} + 2J < 0.000361383965535$. The reverse endpoint width is below 0.000363339660948. Thus both implementations pass both frozen targets.

Divide the exact interval endpoints by positive α to obtain $\mathcal{R}$ in inverse-energy units. The physical Euclidean-time integral is $\hbar\mathcal{R}$. No susceptibility is identified. All four exact intervals contain $1/12$, so the interaction-induced inverse shift remains unresolved.

4.5 Controls and review

The producer controls retain zero-time centering, all nodes and endpoint half-weights, a fixed coupling and clock, every arithmetic cost, and actual source provenance. Missing or coherently erased error packets are rejected. A slow atom $(1/4)\delta_{1/16}$ demonstrates the need for an infinite tail within the declared gap/moment-upper class. A mixture $(1/4 - \delta)\delta_3 + \delta\delta_{1/16}$ with $\delta = \epsilon/2$ stays within the uniform proxy tolerance but defeats a free-endpoint-only tail. These are abstract controls, not AQ realizations. The skeptical control $(\delta_{1/16} + \delta_{95/16})/8$ additionally preserves the exact free mass and first moment while retaining a damaging slow tail.

Reversing the quadrature sign excludes the known free inverse. A zero-energy atom gives a constant correlation and divergent inverse, so the centered-gap interface rejects it. The old coupling cap and old cutoff remain insufficient under their complete budgets. These failures are not lower bounds on actual errors.

The independent model-agent review passes 12,373 exact checks, including all 8,194 rows across the two implementations. Normal and optimized reviewer outputs agree; four fresh producer replays reproduce their declared files byte for byte. The producers retain 19 forward and 21 reverse grouped controls. These checks audit arithmetic and provenance; the continuous and infinite-volume arguments supply the mathematical claim. Full derivations are [AT6 forward](#), [AT6 reverse](#), and [AT6 skeptical review](#).

5 Historical lenses and current-source comparison

The named experts are contemporary research lenses supported by selected texts. They did not participate in this project. The source pass is bounded; it is not an examination of every work, patent, government program or subreddit. Historical assertions supply questions and controls, never a Hamiltonian term or a modern theorem premise.

Newton’s December 1678 chymical entries retain weights, residues and unsuccessful mixtures. Selected propositions of his vegetation tract compare a proposed hidden agency with dissimilar processes; his prophetic-language chapter states an interpretive dictionary. These passages motivated explicit error balances, failure-of-analogy controls and physical/model dictionaries. They were read in scholarly transcriptions, not checked against the complete manuscripts [10, 11, 12].

Tesla’s US645576A specification identifies a generator, transmitting and receiving apparatus, a medium and synchronization. Its performance descriptions are patent claims. Vivekananda’s letter of 13 February 1896 reports Tesla’s interest in Vedantic ideas and an anticipated calculation; the inspected passage contains no completed mathematical demonstration. Their useful modern translation is whole-mechanism accounting and a distinction between a proposal and a verified prediction [13, 14]. No occult energy source or consciousness variable is introduced.

Selected Jung passages discuss remembered material that can appear original and changing accounts of a reported dream. This motivated traceable attribution and centering controls. ETH and CERN archive descriptions establish locations for Pauli correspondence; an attempted letter retrieval was blocked and its contents remain unread. A CIA remote-viewing evaluation lead redirected to a generic homepage and was not read. Government custody, archive descriptions and search snippets are not experimental confirmation.

Penrose’s opening spin-network discussion separates a defined combinatorial model from interpretations of its probabilities and geometry. Feynman’s 1974 address emphasizes reporting details that could invalidate a conclusion, alongside personal accounts of investigating mystical claims. These supplied methodological questions; none supplied a gauge-theory proof [15, 16, 17]. The selected passages, exact editions, retrieval limits and technical-source reading depths are recorded in the [modern](#), [Newton/Tesla](#), and [Jung/Pauli source ledgers](#).

The nearest inspected modern techniques are local product-reference perturbation bounds, Duhamel comparison, Poisson spectral calculus and certified arithmetic. Henheik–Teufel–Wessel’s local stability theorem [5] permits unbounded on-site terms, but its stated existential constants do not numerically certify the AQ cap. The 2026 causal-bootstrap and Mutzel–Tilloy papers [6, 7] constrain spectral functionals from input Euclidean data; they do not produce this model’s input data. Their statistical covariance regions are distinct from the deterministic errors here. Tsuji–Hashimoto’s SVD approach [8] addresses inverse-Laplace conditioning. Li’s August 2026 interaction-tail framework [9] assumes suitable gapped paths and is not automatically a spatial-volume theorem for AQ. The inspected neural SU(2) model with dynamical fermions is a different 2 + 1-dimensional variational problem. No theorem or numerical output from those different models is silently transferred.

These established methods retain their attribution. The contribution labels in this addendum identify the project’s particular model dictionary, constants, evaluated enclosures and falsifying controls. Scientific priority has not been established by an exhaustive comparison.

6 What the certificates do and what remains open

The practical output is an explicitly bounded comparison for an actual, infinite-volume AQ state in a small nonzero-coupling patterned subfamily. The reference curve is calculable and the interval pays the difference from that reference; this does not turn the reference into the exact interacting answer. A common enclosing interval for different allowed subsequential states is not a proof that those states coincide. Deterministic analytic proxies are not independent observations, and their errors cannot be divided by the square root of the number of nodes.

The direct proof avoids a quantitative finite-volume error modulus for its own analytic enclosure: local reset and dynamics estimates hold uniformly before the already admitted AQ limit. It does not certify an arbitrary finite-box solver. Such a solver still needs state/boundary comparison, complete representation-cutoff and omitted-channel bounds, actual ground projection and energy errors, evolution error, centering error, arithmetic error and the specified physical clock. Observed stability of digits does not supply these missing hypotheses.

The unchanged fundamental limits are substantial: no uniform Wilson magnetic Hamiltonian is proved by the zero-selected calculation; no all-state uniqueness, AO/AQ identification, quantitative boundary independence, resolved particle pole, dispersion law, susceptibility or matched continuum theory follows. An inverse-energy form is mathematically defined on the vacuum-orthogonal sector; identifying it with a thermodynamic response derivative would require a separately specified deformation and differentiability theorem. The Clay problem requires the four-dimensional continuum construction and its physical properties, not only a fixed-spacing estimate [18].

The updated [prospective roadmap](#) orders the next questions by what this work does not establish:

1. Return to $|\tau| = 10^{-8}$ with a sharper local state or observable estimate. A creation-expansion approach must pay its complete normalization, support and remainder costs; a formal leading term is insufficient.
2. Resolve an interaction-induced effect by an interval that excludes its matched free value, with the observable and separation target fixed in advance.
3. Match selected and omitted coefficients to a uniform Hamiltonian, charging selected interactions or proving a genuinely non-Haar reference comparison. Even a uniform strong-coupling result is not a continuum limit.
4. Compare declared boundary and state constructions quantitatively in the required local topology and at a common physical clock.
5. Establish a regulator/coupling trajectory, scale control and the continuum reconstruction hypotheses, or isolate a specific necessary estimate that fails along the chosen route.

Every item is unexecuted. No fraction of the final continuum proof is inferred from the number of

loops, equations, pages or named contributions. The existing historical-panel, Newton, Tesla and paired-research skills were retained; future investigations require fresh prospective contracts and source snapshots.

A Contribution and evidence ledger

The following names are internal attribution aliases for this project. Established methods retain their existing authorship; no alias establishes scientific priority.

Project alias	Scoped contribution	Boundary and prior method
HNM-C-AT4	Seven-star actual-state comparison in the zero-selected AQ corner; evaluated cap-level error and equal-moment obstruction.	Target limited. Uses established trace distance, bounded Duhamel, Poisson calculus and moment-information methods. Abstract A/B measures do not prove AQ nonuniqueness.
HNM-C-AT5	An exported exact datum and full radius at a prospectively changed, nonzero coupling; reusable scoped calculators.	Accepted within scope. Evaluates the admitted AT4 method; no interaction shift or original-cap precision is resolved.
HNM-C-AT6	Continuous actual-state window, all 4097 certified rational proxies, and two evaluated inverse-energy enclosures.	Accepted within scope. Uses established positive spectral tails and Peano quadrature. The endpoint candidate was shared before production; no exact interacting trajectory or susceptibility follows.

The forward and reverse workers derived and implemented each frozen question separately before exchanging current solutions. They shared the AQ/AT premises, the prospective reset/Duhamel/Poisson route, and later the reviewed AT4/AT5 methods. The forward positive-effect refinement was separately checked after AT4 freeze. Separate skeptic roles derived controls before comparing the producer outputs. All roles are model-agent work with correlated ancestry, not independent physical observations or external human peer review. AT6 worker-session continuations completed saved checkpoints; they count as neither new investigations nor additional independent reviews.

The immutable decision records are the [AT4 gate](#), [AT5 gate](#), and [AT6 gate](#). Each binds its contract, frozen reports, executable checks, source snapshots, outputs and review evidence. Fresh normal/optimized replay validates exact output reproducibility and source binding. Gate-specific snapshots preserve the actual instructions used; changing later guidance cannot rewrite an earlier admission.

This addendum’s builder refuses a completed build without all three gates and verifies their source digests. Its build receipt binds the manuscript and scientific-source inventory to the PDF. The separate QA record binds geometry, reference resolution and actual review of every rendered page to the PDF’s digest. Reproduce and check the manuscript from the repository root with

```
python -B papers/round31-addendum/build.py -render
python -B papers/round31-addendum/build.py -check
```

The source-bound research replay and staged release checks remain distinct from visual PDF review. Neither a hash nor a passing program replaces the stated mathematical hypotheses.

Exactly three investigations are closed. Literature reading, repairs, replay, manuscript work and publication add no research loops. The prospective goals remain unexecuted. Draft03 and all earlier scientific verdicts are preserved.

References

- [1] Hruday N M (BUNZEEY), with model-agent assistance, *Yang–Mills workbench, Round29 AQ1/AQ2 and Round30 AT1–AT3*, reviewed project evidence, 2026. [AQ checkpoint](#); [Round30 checkpoint](#). These are inherited project premises, not external human peer review.
- [2] S. Gauvin, *Pinched Multi Affine Geometry and Confinement: Describing the Yang–Mills Mass Gap*, arXiv:2503.15539v3, 14 September 2026, Supplemental Material, A.10, pp.S6–S8. <https://arxiv.org/abs/2503.15539v3>. Prior compactness/GNS strategy; reading and model-specific limits are recorded in the inherited AQ source dictionary. No endorsement of the source’s complete claims is implied.
- [3] NIST Digital Library of Mathematical Functions, §1.15(v), *Poisson’s integral*, Eqs.1.15.33–1.15.37. <https://dlmf.nist.gov/1.15#v>. Advisor source check: kernel normalization and absolute-frequency multiplier.
- [4] B. Nachtergaele and R. Sims, *On the dynamics of lattice systems with unbounded on-site terms in the Hamiltonian*, arXiv:1410.8174v1 (2014). <https://arxiv.org/abs/1410.8174v1>. The actual dynamics dictionary is inherited through AQ1.
- [5] J. Henheik, S. Teufel and T. Wessel, *Local stability of ground states in locally gapped and weakly interacting quantum spin systems*, Lett. Math. Phys. 112, 9 (2022), arXiv:2106.13780v3, 3 March 2022. <https://arxiv.org/abs/2106.13780v3>. Read: Section 2 assumptions, Theorem 3 and selected boundary discussion.
- [6] R. Abbott, S. Fields, W. I. Jay, P. Oare and M. Saccardi, *The Causal Bootstrap: Bounding Smeared Spectral Functions from Non-Perturbative Euclidean Data*, arXiv:2605.20509v1, 19 May 2026. <https://arxiv.org/abs/2605.20509v1>. Read: selected exact-data, duality and extremal-measure passages.
- [7] S. Mutzel and A. Tilloy, *Certified spectral functions from lattice Monte Carlo data*, arXiv:2606.09791v1, 8 June 2026. <https://arxiv.org/abs/2606.09791v1>. Read: Sections II.1–III.1 and beginning of III.2. The rendered body also displays 24 August; the date above is the submission-history date.
- [8] R. Tsuji and S. Hashimoto, *Spectral reconstruction from Euclidean lattice correlators through singular value decomposition*, arXiv:2605.15674v1, 15 May 2026. <https://arxiv.org/abs/2605.15674v1>. Read: abstract and opening technical formulation only.
- [9] K. Li, *Local observable errors from truncating interaction tails in gapped quantum lattice systems*, arXiv:2608.15576v1, 16 August 2026. <https://arxiv.org/abs/2608.15576v1>. Read: selected introduction, summary and locality/boundary setup; full proof not audited.
- [10] I. Newton, *Experiments*, Portsmouth Add. MS.3973, entries 10–17 December 1678, ff.1r–1v, Indiana University Chymistry of Isaac Newton scholarly transcription. <https://newton.dlib.indiana.edu/text/ALCH00109/normalized>.
- [11] I. Newton, *Of Natures obvious laws & processes in vegetation*, Dibner MS.1031 B, selected ff.1r–1v, Indiana University scholarly transcription. <https://newton.dlib.indiana.edu/text/ALCH00081/normalized>.
- [12] I. Newton, *Observations upon the Prophecies of Daniel*, Part I, Chapter II (1733), selected opening interpretive mappings, Oxford Newton Project transcription. <https://www.newtonproject.ox.ac.uk/view/texts/normalized/THEM00196>.
- [13] N. Tesla, *System of transmission of electrical energy*, US645576A, filed 2 September 1897, granted 20 March 1900. Selected specification and claims read through patent OCR. <https://patents.google.com/patent/US645576A/en>.
- [14] S. Vivekananda, letter to E. T. Sturdy, 13 February 1896, *Complete Works*, vol. 5, selected Tesla-related paragraphs. [Complete Works transcription](#). This is a correspondent’s report, not a Tesla-authored calculation.

- [15] C. G. Jung, *Collected Papers on Analytical Psychology*, ed. C. E. Long, authorised English translation, 1920 second-edition reprint, selected pp.85–89 and 178–185. <https://www.gutenberg.org/files/48225/48225-h/48225-h>.
- [16] R. Penrose, *Angular momentum: an approach to combinatorial space-time*, in *Quantum Theory and Beyond*, ed. T. Bastin (1971), pp.151–180. Read: opening 13 pages of the academic-hosted transcription. <https://math.ucr.edu/home/baez/penrose/Penrose-AngularMomentum.pdf>.
- [17] R. P. Feynman, *Cargo Cult Science*, Caltech commencement address (1974). Read: selected opening and scientific-integrity passages. <https://calteches.library.caltech.edu/51/2/CargoCult.htm>.
- [18] A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, official Clay Mathematics Institute problem statement. <https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf>. The continuum problem is distinct from the fixed-spacing model studied here.